Structural and Influence Analysis of Financial Systems

S&P 500 Correlation Network

Angadjeet (2022071) Aniket Gupta (2022073) Parth Rastogi (2022352)

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Outline

1. Network Construction
2. Scale-Free Analysis
3. Contagion
4. Event Validation
5. Community Structure
6. Temporal Dynamics

Network Construction

Problem:

Raw Pearson correlations are dominated by the common market factor, producing a dense, uninformative graph.

Methodology:

- 2-year daily log-returns for $N = 501$ stocks.
- **Market-mode filtering** (Onnela et al. 2003): subtracted equal-weight market index.
- **Thresholding**: edges where $|\rho| > 0.3$.

Result: Mean $|\rho|$ fell from $0.235 \to 0.127$; the filtered graph has 10,002 edges.

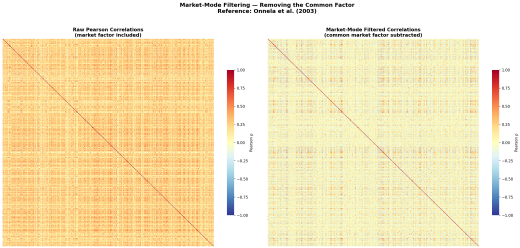


Figure 1: Raw vs. filtered correlation heatmap.

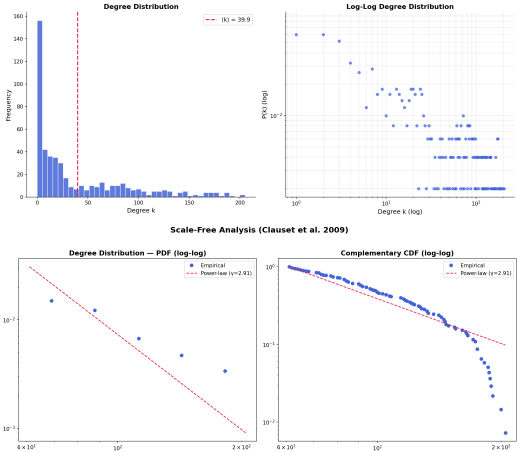
Degree Distribution

Connectivity

- 47 connected components; Giant Component holds 89.4% of all nodes.

Power-Law Fit (Clauset et al. 2009)

- Heavy-tailed degree distribution.
- $P(k) \sim k^{-\gamma}, \quad \gamma \approx 2.14.$
- Confirms hub-dominated, scale-free topology.



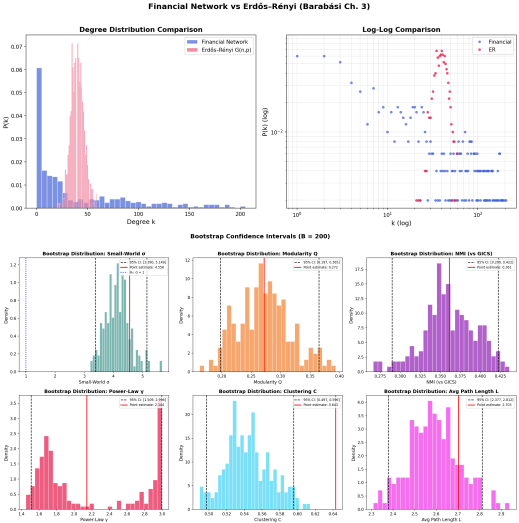
Small-World

ER Baseline

- Against Erdős–Rényi: $\sigma_{ER} = 4.56$.
- ER underestimates clustering due to ignoring degree heterogeneity.

Configuration Model

- Degree-preserving null model gives $\sigma_{config} = 1.75$; still > 1.
- Bootstrap ($B = 200$): $p < 0.001$, confirming significance.
- Weighted clustering $C^w = 0.27$ vs. unweighted $C = 0.64$ (Barrat et al. 2004).



Influence Propagation

Cascade Model

- Degree centrality vs. cascade spread: $r = -0.61$. High-degree nodes dilute per-edge impact through row-normalised weights.

Robustness (Barabási Ch. 8)

- Random failure: GCC degrades gracefully (scale-free resilience).
- Targeted betweenness attack: GCC collapses rapidly.

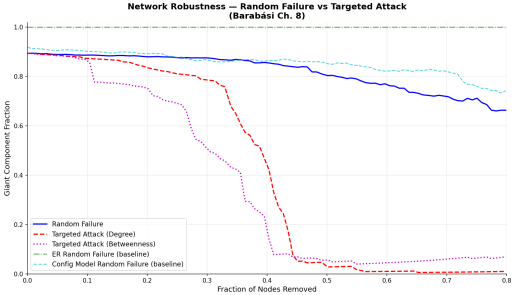


Figure 2: Robustness under random vs. targeted attack.

Event Validation

Approach

Rather than synthetic shocks, we seed propagation with actual event-day returns and compare model predictions against realised post-event performance.

Scenarios:

- AI / Semiconductor Rally (Jul 2024)
- Yen Carry-Trade Unwind (Aug 2024)
- Tariff Shock (Apr 2025)

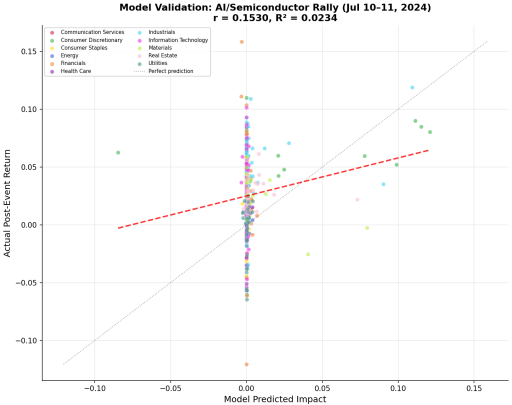


Figure 3: Model vs. actual 5-day returns.

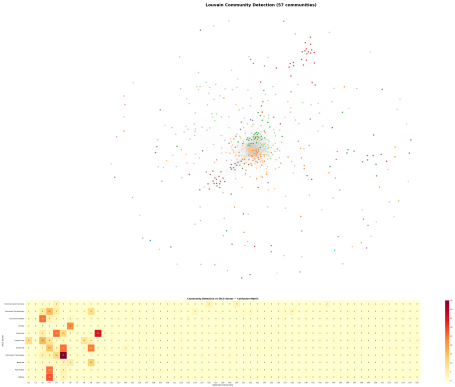
Community Detection

Louvain Partition

- 57 communities, modularity $Q = 0.27$.

Alignment with GICS

- $NMI = 0.45$: partial overlap.
- Network communities capture cross-sector supply-chain linkages that official GICS sectors do not.



Rolling Windows

Setup

- 18 overlapping 6-month windows spanning the full 2-year period.

Findings

- Clustering and modularity increase during stress periods.
- The network tightens when macro shocks hit—consistent with "flight to correlation" behaviour.

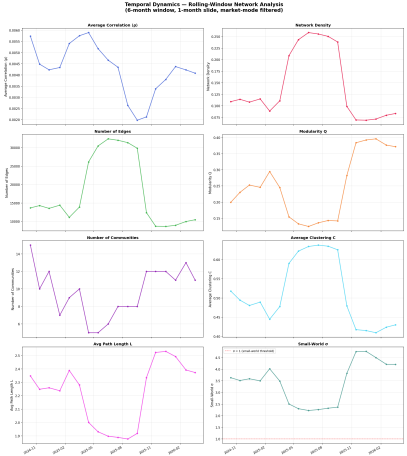


Figure 4: Structural metrics over rolling windows.

Conclusions

Key Finding

The S&P 500 correlation network is scale-free, small-world, and highly modular—yet systematically vulnerable to targeted attacks on high-betweenness nodes.

Contributions:

- **Filtering**: Market-mode subtraction removes spurious correlations.
- **Null Models**: Configuration model confirms small-worldness beyond degree effects.
- **Validation**: Propagation model tested against three real market events.

Thank You.